Generalization From Relevance Feedback in Content-Based Image Retrieval

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Abstract

Many recent content-based image retrieval techniques utilize relevance feedback (RF) from the user to adjust the system response to better meet user expectations. One school of RF-based methods uses a weighted Minkowski distance metric to assess similarity, and adjusts the weights to refine query response. Such methods usually fail to generalize from previous user interactions and require each RF session to begin from a naive initial weight vector. A new, task-based method is proposed which uses RF to adjust the weightings used in a Minkowski distance metric, and which uses all previous user interactions on a task to produce initial estimates for new target images never before encountered by the system.

1. Introduction

Content-Based Image Retrieval (CBIR) techniques typically represent each database image as a feature vector in an N-dimensional Hilbert space. In query-by-example type searches, users provide a new feature vector representing the query image. The system then returns the images closest to the query vector in the feature space, where closeness is defined by a similarity measure $d(\mathbf{x}, \mathbf{y})$. Relevance feedback (RF)-driven systems present query results to the user and receive information back from the user indicating which query images are judged relevant, and which are not. RF-driven systems may also request that the user order the relevant images.

1.1. Previous Work

The methods used to implement relevance feedback usually fall into three broad categories: *query refinement*, *metric refinement*, and *probabilistic*. Query refinement type methods refine a model of the "ideal" query vector with each iteration based on the distribution of the labeled database images. Metric refinement type methods use flexible similarity measures, such as weighted Minkowski metrics [1, 2, 3, 4]. These measures can adjust the importance to the query of each dimension, or feature, according to the feedback from the user. (Query refinement and metric refinement type feedback operations are not mutually exclusive, and have been shown to be synergistic with each other. [2]) Probabilistic approaches use a model of the user to determine which images should be presented as a response to a query image, estimating the probability of each database image being the target image.

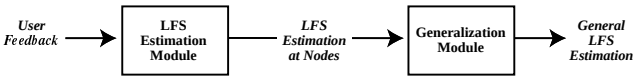


Figure 1: Two Primary System Modules

For reasons which will become clear later, this work focuses on the metric refinement type RF methods. Existing metric refinement type techniques have not adequately addressed the initial value estimation of the weight vector. This implies that each new query session must start with a naive weight choice, typically the choice of equal weights: $w_k = 1/N$. This assumption places a heavy burden on the system operator, as a significant number of feedback interactions is required to find the best weights for the system.

2. Architecture

This work seeks to mitigate the demands placed on system users in the day-to-day operation of a CBIR system on a well-defined domain. In order to accomplish this, a two-tier architecture is proposed, as presented in Figure 1. The first system module is responsible for estimating the pertinence of each feature for a query on the provided image, given user feedback associated with the image. The pertinence, or saliency, or each feature is then reflected by the weight used by the distance measure. It is known that the saliencies of each feature for a query are generally not constant across the domain of images in a database [1]. This work therefore treats every set of feature saliencies as explicitly tied to a particular query image, and therefore a particular point in the feature space. They are referred to as the *local feature saliency* (LFS) values at the corresponding point. The first system module is referred to as the LFS estimation

module. Most methods proposed in the literature are the performance equivalent of this module only.

The second system module is the generalization module, and is responsible for providing LFS estimates at points in the feature space which have no associated user feedback. This corresponds to the use case of the presentation of a novel image to the RF-driven CBIR system. This, of course, is the most common course of action undertaken by a CBIR system in the field. In the proposed architecture, the generalization module takes from the first module the collective set of LFS estimates produced from all feedback specific to a given task and uses them to provide LFS estimates at any point in the space. Note that the system partitions the set of all user feedback by *task* - a reasonable step to make as different tasks will result in different saliency values even on the same query image. Many typical fielded CBIR systems involve performing the same task repeatedly on a well-defined image domain, and the current work targets this problem.

2.1. Local Feature Saliency Estimation

A new method of estimating preferred weights for use in the weighted Minkowski distance metric was introduced by this research effort, and is described in detail elsewhere [5] [6]. This method sought to define an objective function for the weight optimization procedure which more closely represented the true goals of the system: to rank the query images in a manner consistent with that expected by the human user. The result of this method is the production of an optimized weight vector for each query image using all available user feedback associated with the image.

In summary, this method considers the relationships $\lambda_{ij} = s_j - s_i$ between the scores of images considered relevant to a query image (indexed by $i \in \mathcal{R}$, the set of indices of relevant images) and those not considered relevant (indexed by $j \in \mathcal{N}$, the set of indices of non-relevant images). For a (pseudo-)metric based query operation, it is desirable that $\lambda_{ij} > 0$. The weight optimization routine employs a penalty function, $\psi(\cdot)$, which penalizes when this inequality is violated. This estimation method can also incorporate relative judgments using the relationships λ_{mn}, which represent the desired ordering of relative images, where $m \in \mathcal{R}$, $n \in \mathcal{R}_m$ and $\mathcal{R}_m \subset \mathcal{R}$ is the set of images similar to the query image, but less similar than image m.

Explicitly acknowledging the dependency of the scores, $s_i(\mathbf{w})$, on the weight vector $\mathbf{w}$, with $w_i \geq 0$, the goal is then to determine the feature weights, $\mathbf{w^{opt}}$, that minimize

$$\mathbf{w^{opt}} = \arg\min_{\mathbf{w}} \left[\alpha \sum_{i\in\mathcal{R}} \sum_{j\in\mathcal{N}} \psi(\lambda_{ij}(\mathbf{w})) + \beta \sum_{m\in\mathcal{R}} \sum_{n\in\mathcal{R}_m} \psi(\lambda_{mn}(\mathbf{w}))\right]$$

This optimization problem can be approached in a number of ways, including the use of a differentiable penalty function $\psi(\cdot)$ to facilitate a gradient descent-type solution, or the use of a support vector machine framework to find maximally marginal solutions. For details, see references [5, 6], which demonstrate the superiority of this framework over previous frameworks.

2.2. Generalization

The production of a useful set of LFS estimates enables the completion of a query requested by the user. When a new query image is provided, the LFS estimation process begins anew, with user feedback iterations refining a naive initial estimate of the saliencies. The current work proposes a method of utilizing all previous feedback iterations associated with a user task to to provide an improved initial weight vector for the user interaction process. In a stable domain, this ability to predict the LFS values of new images can significantly reduce the daily workload of system users, and in the best case eliminate the need for user feedback on a trained system altogether. A mapping function which outputs an LFS estimate given a location in the feature space is referred to as a *local feature saliency map*. The ability to maintain such a mapping is one of the primary benefits of using a metric-refinement type technique, rather than a query-refinement technique, which models the ideal query vector sought by the user for the query at hand.

. Requirements

A variety of solutions could be used to provide LFS estimates for all points in the feature space. The following requirements were adopted to guide the choice of method:

1. Output weights at the location of nodes should be very close to that which would be given using only LFS estimates at that node
2. Output weights should transition smoothly between nodes

The first requirement states that the generalization process should not distort the LFS estimates at the node points themselves. While some distortion may be necessary as a trade-off for other generalization goals or computational complexity, this should be kept to a minimum, as the estimates from the LFS module are the best information available. The second requirement ensures smooth transition of weights between nodes. This requirement is based on the important assumption that images close to each other in the feature space should have similar desired weight values. This assumption is explored below.

. Proposed Solution

The requirements defined above largely correspond to the general interpolation problem. This problem has been well studied over many years, and a large number of solutions are available. Perhaps the most commonly used approach is the spline-fitting solution, which fits piecewise functions to

the nodes while minimizing smoothness constraints. Note, however, that the input data to the generalization process are not guaranteed, nor even likely, to have the regular spacing assumed by typical spine-fitting processes. The fact that the irregularly spaced data is in a high dimensional space further compounds the problem.

Radial basis function (RBF) networks are a type of feed-forward neural network that has gained considerable attention in recent years. The purpose of a RBF network is to provide a mapping function $F : \Re^N \rightarrow \Re^K$, which maps an input vector $\mathbf{x} \in \Re^N$ to an output vector $\mathbf{y} \in \Re^K$. The non-linear activation function can take a variety of forms, but the Gaussian form has several analytical advantages and is most commonly used. RBF networks are well suited for the problem at hand, as they can handle irregularly sampled training data in high-dimensional spaces.

One of the primary drawbacks of radial basis functions, however, is their susceptibility to performance degradation in the presence of irrelevant features. This is an incarnation of the infamous "curse of dimensionality", where the number of bases required to provide support over the space increases exponentially with the number of features. One approach to dealing with irrelevant features is to use *weighted norms* in the implementation of the RBF. In this case, rather than using a spherical Gaussian basis, an elliptical Gaussian basis can be used to allow the specification of the basis' support along each feature. Irrelevant features should have broad support by bases, while relevant features warrant narrower support. The weighted norm problem then becomes an issue of determining the relevance of each feature and weighting the Gaussian accordingly. To this end, the current framework offers an interesting opportunity to exploit the training data for multiple purposes. Each component of an output training vector reflects the relevance, or saliency, of that component at the corresponding point in the input space. This information can used to shape the weighted norm basis function in question by making the component weights of the basis proportional to the output LFS weights provided at that node. Therefore, the bases are defined as

$$\phi_j(\mathbf{x}) = \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mathbf{x}^j\right)^{\mathbf{T}}\mathbf{W}_j\left(\mathbf{x}-\mathbf{x}^j\right)\right)$$

where

$$\mathbf{W}_j = diag\left(\mathbf{w}_j\right) \propto diag\left(\mathbf{t}^j\right)$$

and $(\mathbf{x}^n, \mathbf{t}^n)$ is a pair of input and output training vectors.

3. Experimental Results

This section presents performance results of the generalization module. A database of 370 images was assembled of Tennessee wildflowers, 343 of which had at least one similar image in the collection (typically 2 or 3). This database composition is designed to provide a large number of query images with limited training data for similarity searches, a common real-world CBIR scenario. Color histograms of 27 bins were extracted as features for each image. The "Goldilocks principle" of analysis of learning systems[1] was followed in choice of task, database, and feature set. To quantitatively assess the performance, a query performance measure (QPM) compared the system output to predeclared human judgments, and penalties were assessed for false positives, false negatives, and transpositions.

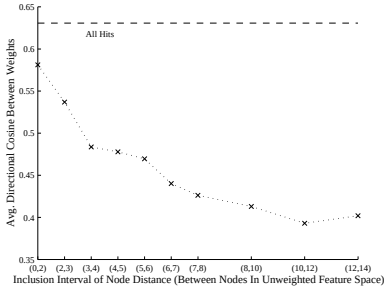


Figure 2: Average Directional Cosines Between Weights

3.1. Plausibility

A plausibility study was conducted to determine if nodes which are closer together have optimum weight vectors that are more similar than nodes further apart. The collection of 343 nodes and optimum weight vectors, as determined by the LFS module, was considered for this study. Each pair of distinct nodes was analyzed for two quantities: the unweighted Euclidean distance between the nodes in the feature space, and the similarity between the associated weight vectors, as measured by the directional cosine measure, s_{dc}. The node pairs were assigned to one of 10 bins according to the distance between the nodes, and the average s_{dc} between the weight vector pairs is calculated for each bin. The resulting averages are plotted in Figure 2. It is very clear from the plot that as images are closer together, even in the unweighted feature space, it is more likely that the weight vectors are also similar. This evidence provides some justification for the approach of the generalization module of using nearby known weight values in estimating unknown weight values at a new point in the space.

3.2. Generalization Capability

The generalization capability, or *persistence*, of the system was tested using a k-fold cross-validation procedure. For these tests, $k = 10$ was used as a trade-off between the bias of the error estimate and the computational cost. This test compared the performance of the trained system to the performance of an untrained, or naive system. The naive system represents the performance which could be

[1] To avoid ceiling and floor effects, the task must neither be "too easy" nor "too hard", but rather "just right".

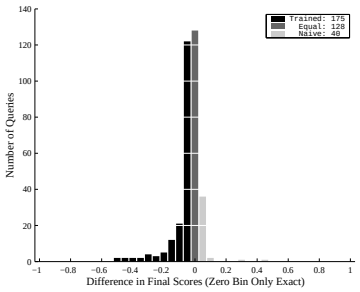


Figure 3: Difference Histogram: Generalization Performance

expected from a CBIR system without any generalization ability. Note that this level of performance does not represent the final possible performance of either the present CBIR system or another relevance feedback-based CBIR system, but rather the starting point before any feedback is given on the new image. Persistence reduces, and in the best case could eliminate the requirements for feedback from the user. Without persistence, a CBIR system must start from the naive state with every new image. This is one of the central advantages of this research.

The results of the test are shown in Figure 3. In 175 of the 343 query images, the system was able to use its previous experience with the user on other images to improve its estimate of the LFS for the new image. In 40 of the 343 images, the system's estimate was worse than the naive system. The superiority of the trained system showed a strong statistical significance ($p < 0.0001$).

The relatively small percentage of images which performed worse with the new system is not problematic. Expectations cannot be set unreasonably high: a system that always was able to guess every new image might be called a "psychic system", the cousin of the perpetual motion machine. Except in the best case of large collections of images with stable tasks and consistent users, some new images which require LFS estimates unlike those previously encountered by the system are expected.

Another test compared the performance of the trained system on and off the training nodes. The first mode of operation was trained using the same k-fold cross-validation, using a training set and a test set. The second mode tested images which had been encourtered before, therefore using no generalization. The practice of using the same data for training and testing is sometimes referred to as the *resubstitution* method in the statistical pattern recognition literature, and is used to provide an upper bound on the performance. Not surprisingly, the resubstitution method outperformed the generalizing system in almost every case. The mean QPM values of the three methods is provided in Table 1. The naive system and the resubstitution test bracket the performance of the trained system used in normal operation.

Table 1: Mean QPM Values from Generalization Tests

Method	Mean QPM
Naive System	0.16662
Trained System	0.13476
Resubstitution	0.05409

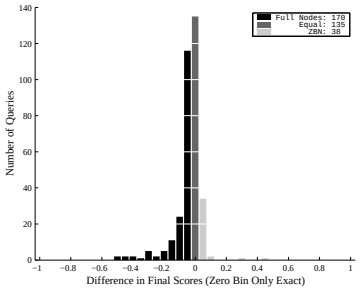


Figure 4: Difference Histogram: Local Feature Saliency vs. Global Feature Saliency

3.3. *Local* Feature Saliency

Another focus of this research is the introduction of local feature saliency maps for better specification of the LFS values as a function of the position of the image in the feature space. Without LFS, every image is treated the same by the CBIR system. The effectiveness of LFS maps was studied using a k-fold cross-validation test. The system was operated with and without the LFS capabilities. The crippled system is referred to as a *zero basis net* (ZBN) implementation. The regular implementation and the ZBN implementation were tested using the same parameters for the LFS modules and the same user feedback was provided as input to the LFS module. The results of the test are shown in Figure 4. The system overwhelmingly performed better when the LFS maps were used. In 170 of 343 queries, the LFS-enabled system outperformed the ZBN system. In only 38 cases did the ZBN outperform the regular system. Statistically, this result is strongly significant ($p < 0.0001$). This test clearly shows that substantial benefits are gained by maintaining an LFS map.

The poor performance of the ZBN system begs the question of whether the ZBN provides any benefit at all. For insight into this question, the ZBN was compared to the naive system. Both systems use a single weight vector across the entire space. The naive system uses the initial weight vector of equal weights. The ZBN system adjusts the weight vector based on the feedback available, and sets it as the average of all weight vectors provided from the LFS module. The systems were compared in three scenarios, summarized in Table 2. The first scenario used only one feedback set for the training, therefore only 34 or 35 images are used for the training. The ZBN system improved the system over the naive guess with a high degree of significance. However, when all 343 query images are included in the training in

Table 2: ZBN Capacity Tests

Trained on Test Set	Feedback Sets	p (Naive Better)
yes	$1/10$	$p < 0.0003$
yes	$10/10$	$p < 0.5064$
no	$9/10$	$p < 0.5833$

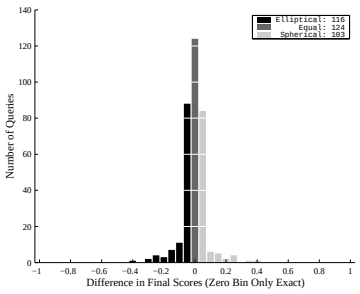


Figure 5: Difference Histogram: Elliptical Bases vs. Spherical Bases

the second scenario, the ZBN performs slightly worse than the naive system. This comparison is statistically insignificant. The third scenario operates the system in the normal operation mode, where the training does not occur on the test image, and the number of images in the system is high. Again, in this case the naive system outperformed the ZBN system, but the difference was not statistically significant. None of these performance gains or losses were substantial in magnitude.

These tests seem to indicate a capacity limit in the ability of a ZBN to perform adequately. When a small number of images are included in the training set, the ZBN provides some advantage. This advantage dissipates when the training set size increases, perhaps because the weights necessary for good performance vary too much over the set. It appears that a fundamental limitation, or performance bias, exists on the ability of the ZBN system to do well on this data set when a large number of optimum weight vectors must be balanced against each other. This observation also argues for the importance of local feature saliency estimations.

3.4. Shaped Bases

Another test considered the effectiveness of the shaped bases vis-a-vis spherical bases. The results of the test are displayed in Figure 5. The elliptical bases were seen to offer a modest improvement against the circular bases. In 116 of the 343 queries, the elliptical bases performed better. In 103 of the queries the spherical bases performed better. The tail of the distribution on the left is slightly fuller than the tail on the right, which contributed to the mean value of the final QPM being somewhat lower, as shown in Table 3. This improvement, though consistent in many experiments, was not seen to be statistically significant ($p < 0.1399$).

Table 3: Mean QPM Values of Shaped Bases Tests

Method	Mean QPM
Elliptical Bases	0.13476
Spherical Bases	0.13906

Table 4: Mean Number of Activated Bases

Method	$t_a = 0.05$	$t_a = 0.25$	$t_a = 0.5$
Elliptical Bases	5.55	1.91	0.70
Spherical Bases	77.11	20.34	4.23

The modest improvement offered by using shaped bases belies a significant difference in the internal operation of the two types of networks. Table 4 shows the average number of bases activated for different levels at the best respective σ values. The network using shaped bases clearly makes more effective use of its bases. The fact that the two networks are operating so differently internally suggests that this area holds further potential as a research topic.

4. Discussion

A new method of using RF in CBIR was presented which improves the ability of the system to make educated initial guesses for retrieval parameters. This method was seen to provide tangible benefits to the system user with improved query responses before any interaction is required. In a stable domain, the expected reduction in the amount of feedback iterations required will streamline the workflow, and increase acceptance of the system. The benefits of using local feature saliencies were also demonstrated, as were the limitations of using global feature saliencies. The use of estimated LFS values for shaping RBF bases was seen to offer modest improvement, but this research avenue has not been exhausted and may yield further benefits still.

References

[1] B. Bhanu *et al.*, "Learning feature relevance and similarity metrics in image databases," in *IEEE CBAIVL*, 1998.

[2] J. Huang *et al.*, "Combining supervised learning with color correlograms for content-based image retrieval," in *Proc. ACM Conference on Multimedia*, p. 325, ACM, 1997.

[3] Y. Ishikawa *et al.*, "Mindreader: Querying databases through multiple examples," in *Proc. 24th VLDB Conf.*, 1998.

[4] Y. Rui *et al.*, "A relevance feedback architecture for content-based multimedia information retrieval systems," in *IEEE CBAIVL*, pp. 82–89, IEEE, 1997.

[5] B. C. Jones, *Local Feature Saliency Maps for Persistent Relevance Feedback-Driven Content-Based Image Retrieval*. Ph.D. dissertation, Vanderbilt Univ., 2001.

[6] B. C. Jones and D. M. Wilkes, "A new analysis framework for relevance feedback-driven similarity measure refinement in content-based image retrieval," in *submitted to CVPR, 2001.*